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MoSDeF-dihedral-fit: A lightweight software for fitting dihedrals within MoSDeF

Brad Crawford [1,2], **Co D. Quach** [3,4], **Nicholas C. Craven** [4,5], **Christopher R. Iacovella** [3,4,5], **Clare McCabe** [3,4], **Peter T. Cummings** [3,4], and **Jeffrey J. Potoff** [2]

1 Atomfold, PA, USA **2** Department of Chemical Engineering and Materials Science, Wayne State University, Detroit, MI 48202-4050, USA **3** Department of Chemical and Biomolecular Engineering, Vanderbilt University, Nashville, TN 37235-1604, USA **4** Multiscale Modeling and Simulation (MuMS) Center, Vanderbilt University, Nashville, TN 37212, USA **5** Interdisciplinary Material Science Program, Vanderbilt University, Nashville, TN 37235-0106, USA

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Software

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Summary

Molecular Mechanics (MM) simulations (e.g., molecular dynamics and Monte Carlo) provide a third method of scientific discovery, adding to the traditional theoretical and experimental scientific methods (Mielke et al., 2019; Siegfried, 2014). Experimental methods measure the data under set conditions (e.g., temperature and pressure), whereas the traditional theoretical methods are based on analytical equations, and sometimes their constants are fitted to experimental data. The MM simulations are deterministic and stochastic, and their models, commonly known as "force fields", can be optimized to match experimental data, similar to analytical theory-based methods (Allen & Tildesley, 2017; Errington & Panagiotopoulos, 1999; Frenkel & Smit, 2002; Hemmen & Gross, 2015; Jorgensen et al., 1996; Martin & Siepmann, 1998; Potoff & Bernard-Brunel, 2009; Scott J. Weiner et al., 1984; S. J. Weiner et al., 1986). In larger, more complex systems, the stochastic simulation's molecules can jump large energy barriers that deterministic simulations may not be able to overcome in a reasonable timeframe, even with modern computing capabilities (Allen & Tildesley, 2017; Frenkel & Smit, 2002). However, deterministic and stochastic systems that provide adequate sampling for calculating a given property can provide critical insights into the system's phase space, which are not obtainable via traditional theoretical and experimental methods. Additionally, molecular simulations provide critical insights from visualizations and by obtaining chemical or material properties that do not currently exist, are not easily attainable (e.g., too expensive or dangerous) by traditional theoretical and experimental methods (Hirst et al., 2014; Hollingsworth & Dror, 2018), or require hard-to-achieve conditions, such as very high pressures and temperatures (Koneru et al., 2022; Kumar et al., 2022; Louie et al., 2021; Swai, 2020; Yu & Pahl, 2023). However, the force field parameters are ideally determined from Quantum Mechanics (QM) simulations or other methods, including the vibrational spectrum and machine learning methods (Friederich et al., 2018; Kania et al., 2021; Mayne et al., 2013; Schmid et al., 2011; Kenno Vanommeslaeghe et al., 2014; Vermeyen et al., 2023). The MM proper dihedrals (i.e., dihedrals) are challenging to obtain if they don't currently exist for the chosen force field, inaccurately scale-up in larger molecules, or misbehave with other moiety combinations, provided some were separately derived using small molecules (Kania et al., 2021; Mayne et al., 2013). While the same QM simulations can fit the dihedrals in most force field types, these dihedrals aren't easily transferable between force fields due to the differing parameters and formulas, including the combining rules and 1-4 scaling factors. (Chen et al., 2015; Huang & Roux, 2013; K. Vanommeslaeghe et al., 2010; Kenno Vanommeslaeghe et al., 2014).

The `MoSDeF-Dihedral-Fit` (Crawford, Quach, et al., 2023) library lets users quickly calculate

the MM dihedrals directly from the QM simulations for several force fields (OPLS, TraPPE, AMBER, Mie, and Exp6) (Errington & Panagiotopoulos, 1999; Hemmen & Gross, 2015; Jorgensen et al., 1996; Martin & Siepmann, 1998; Potoff & Bernard-Brunel, 2009; Scott J. Weiner et al., 1984; S. J. Weiner et al., 1986). The user simply has to generate or use an existing Molecular Simulation Design Framework (MoSDeF) force field XML file (Cummings et al., 2021; *GMSO*, 2019; Summers et al., 2020; Timalsina, 2022), provide Gaussian 16 log or Gaussian-style QM simulation files that cover the dihedral rotation (typically, 0-360 degrees), and provide the molecular structure information in a mol2 format (Frisch et al., 2016). The `MoSDeF-Dihedral-Fit` software uses the QM and MM data to produce the dihedral for the specific force field, fitting the constants for the OPLS dihedral form and then analytically converting them to the Ryckaert-Bellemans (RB)-torsions and the periodic dihedrals. The software outputs the calculated MM dihedral points, enabling users to fit unsupported dihedral forms, provided the force fields are supported by the MoSDeF, GPU Optimized Monte Carlo (GOMC), MoSDeF-GOMC (Crawford et al., 2022; Crawford, Timalsina, et al., 2023; Crawford, Quach, et al., 2023; Nejahi et al., 2019, 2021), and vmd-python (Betz, 2016) software (a derivative of the VMD software (Humphrey et al., 1996; Stone et al., 2001)).

Statement of need

While many of these Molecular Mechanics (MM) force field parameters can be transferred between force fields, such as bonds, angles, and improper dihedrals (impropers), the proper dihedrals (dihedrals) can not be easily transferred due to the different combining rules (arithmetic and geometric) and 1-4 scaling factors (i.e., between the 1st and 4th bonded atoms) that were used in the development of the original parameters (Berthelot, 1898; Good & Hope, 1970; Lorentz, 1881). The accuracy of these dihedral parameters is critical in obtaining the correct molecular conformations and configurations, which are absolutely required for understanding and analyzing the system's microstructure and physical properties (e.g., free energies, viscosities, adsorption loading, diffusion constants, and many more).

Some integrated dihedral fitting software currently exists for AMBER (Horton et al., 2022) or CHARMM-style force fields (Mayne et al., 2013), and other software will fit the dihedral constants to the final MM and QM energies, which need to be calculated by other means (Guvench & MacKerell, 1998). However, there is a need a simple, generalized software package that supports multiple potential functions, imports QM and MM files, automatically reads and organizes the QM data, calculates the MM energies, auto-corrects the dihedral fit to account for multiple instances of the dihedral, and automatically removes the unusable cosine power series combinations due to this symmetry. The `MoSDeF-dihedral-fit` software accomplishes all this and automatically accounts for any of the common combining rules and the 1-4 scaling factors specified via the MoSDeF XML (i.e., force field) files (Cummings et al., 2021; *GMSO*, 2019; Summers et al., 2020; Timalsina, 2022). By allowing the user to set any other dihedral in the molecule to zero, this software avoids forcing one dihedral fit to correct the inaccurate forces of another dihedral, resulting in a problematic or bad cosine series fit; thus, providing a more flexible and accurate fit by combining multiple dihedral conformational energies in a single dihedral, a strategy used in the original and modern OPLS force fields (Jorgensen:1996: Lu et al., 2021). For example, a carboxylic acid with an alkyl tail has two dihedrals in the same rotation cycle; the C-C-C-O: (O: = oxygen without hydrogen) dihedral is set to zero while the C-C-O-H dihedral is fit (Jorgensen et al., 1996; Kamath et al., 2004; Lu et al., 2021). The `MoSDeF-dihedral-fit` (Crawford, Quach, et al., 2023) API fills the missing gap by providing a generalized and easy solution to fitting dihedrals in their commonly used forms and outputting the MM dihedral data points so users can fit other custom dihedral forms.

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Mathematics

Proper dihedral (dihedral) forms.

OPLS-dihedral:

$$U_{OPLS} = \frac{k_0}{2}$$

$$+\frac{k_1}{2} * (1 + cos(\theta)) + \frac{k_2}{2} * (1 - cos(2 * \theta))$$

$$+\frac{k_3}{2} * (1 + cos(3 * \theta)) + \frac{k_4}{2} * (1 - cos(4 * \theta))$$

Ryckaert-Bellemans (RB)-torsions:

$$U_{RB} = C_0$$

$$+C_1 * cos(\psi) + C_2 * cos(\psi)^2$$

$$+C_3 * cos(\psi)^3 + C_4 * cos(\psi)^4$$

$$\psi = \theta - 180^o$$

Periodic-dihedral:

$$U_{Periodic} = K_0 * (1 + cos(n_0 * \theta - d_0))$$

$$+K_1 * (1 + cos(n_1 * \theta - d_1)) + K_2 * (1 + cos(n_2 * \theta - d_2))$$

$$+K_3 * (1 + cos(n_3 * \theta - d_3)) + K_4 * (1 + cos(n_4 * \theta) - d_4)$$

$$where : n_0 = 0; n_1 = 1; n_2 = 2; n_3 = 3; n_4 = 4$$

$$d_0 = 90^o; d_1 = 180^o; d_2 = 0^o; d_3 = 180^o; d_4 = 0^o$$

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